Project Proposal

Smart Library Management System

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Abstract:

The Smart Library Management System is a web-based application developed to automate and manage library operations efficiently. The system replaces traditional manual record-keeping with a digital platform that manages books, students, and issue-return transactions. It provides fast book search, secure data storage, and automatic fine calculation for late returns. The project follows the Waterfall Methodology to ensure systematic development and proper documentation.

Background of the Project

Libraries play an important role in educational institutions, but many libraries still rely on manual processes for managing books and records. These methods are time-consuming and prone to errors. With the advancement of web technologies, library management systems can now be automated. A smart digital system can improve efficiency, accuracy, and data security while providing quick access to information for both students and librarians.

Problem Statement:

The existing manual library system faces several challenges including slow record maintenance, high chances of human error, difficulty in tracking issued and returned books, and inaccurate fine calculation. Additionally, data security is weak, and searching for books consumes significant time. These issues reduce the overall efficiency of library operations.

Solution Overview:

The proposed Smart Library Management System provides a computerized solution to library operations. It automates book and student records, issue-return processes, and fine calculation. The system offers fast book search, accurate tracking, and improved data security through a centralized database, making library management reliable and efficient.

Objective:

To develop a computerized system for efficient library management.

Specific Objectives:
- To reduce manual work and human errors.
- To automate book issue and return processes.
- To provide quick access to library records.
- To calculate fines automatically.
- To improve data security and system reliability.

Scope of the project

The project focuses on developing a web-based library management system for educational institutions. It includes features for book management, student records, issue-return tracking, and fine calculation. The system supports admin and student roles. Future enhancements may include online reservations and mobile application support.

Methodology:

The Waterfall Methodology is used for this project. The development process includes requirement analysis, system design, implementation, testing, and deployment. Each phase is completed sequentially to ensure quality and proper documentation.

Reason for Chosen Methodology

5 to 8 stages of software development are commonly identified by the people. Every model for software development should fulfill these stages requirements to create an acceptable application. If, these stages are not accomplished properly than there are much possibilities for failure of application. These stages are as follows.

- **Requirement Analysis and Definition**
- **System Software Design**
- **Implementation and Unit Testing**
- **Integration and System Testing**
- **Operation and Maintenance**

"Waterfall methodology" is phase developing methodology. Each phase is documented briefly and sends to the user for approval, if it is approved then the next phase is started. If the first phase is not approved by client or it does not according to the requirement of the client it is re-documented.

Because my project requires the same situation so "Waterfall Methodology" is best to use.

Development Tools

Activities	Tools/Programs
Client side coding	HTML,CSS
Client side scripting	JavaScript
Platform	Windows
Database server	MySQL
Server-side scripting	Node.js + Express.js
Browsers	Google chrome
Documentation	MS Word, Adobe reader

Table1

Expected Outcomes:

A fully functional Smart Library Management System.
- Automated book management and issue-return processes.
- Reduced manual workload and errors.
- Fast and efficient book search.
- Accurate and automatic fine calculation.
- Improved data security and reliability.

Conclusion:

The Smart Library Management System aims to replace the traditional manual library system with an efficient digital solution. By automating operations and improving accuracy, the system will enhance library services and provide a better experience for students and administrators.